

IT3012 : Intelligent Agents

Practical 04 — Informed Search & the A* Agent

Completed Answer Sheet — BSc (Hons) in IT Specialized in AI, Year 3 Semester 1

Part 2: Theoretical Evaluation

1. What is the key difference between how UCS and A* Search prioritize which node to explore next?

UCS only looks at $g(n)$, the cost so far, so it expands equally in all directions. A* uses $f(n) = g(n) + h(n)$, adding a heuristic estimate to the goal, so it expands nodes that look closer to the goal first.

2. Why is Manhattan Distance admissible for this 4-way movement grid, and what happens if the heuristic is NOT admissible?

On a 4-way grid, the minimum possible moves to the goal is exactly the Manhattan distance, so it never overestimates the real cost — that makes it admissible. If the heuristic overestimated cost, A* could skip a cheaper path and return a path that is not optimal.

3. If the agent could move diagonally (8-way movement), would Manhattan distance still be admissible? Which metric should you switch to?

No. Diagonal movement lets the agent cover both x and y distance in one step, so Manhattan distance would overestimate the cost. Euclidean (straight-line) distance should be used instead, since it never overestimates the true 8-way movement cost.

4. Propose a stronger heuristic for navigating the grid to eat ALL remaining food efficiently (instead of just the closest food item).

Use the distance to the nearest food plus an estimate of visiting the rest, e.g. the total Minimum Spanning Tree (MST) distance connecting all remaining food. This gives a better idea of the full cost to collect all the food, not just the one closest item.